

Supplementary Material

Learning the Lab, Not the Law: A Verdict-Based Audit Protocol for Structure-Level Domain Confounding in Symbolic Scientific Discovery

This supplement contains material referenced from, but not essential to following the argument of, the main text: extended derivations, the full per-source data-provenance detail, a secondary ablation figure, and three additional analyses added in response to reviewer requests (Sections S4–S6). Every number here is backed by a script or an archived output in the released repository; `run_all.sh` specifically regenerates the S4 and S6 analyses (Sections S4 and S6 below), and none is estimated or asserted without a corresponding script or archived output.

1 S1: Few-cluster inference, extended

Section 3.5 of the main text (Component 3: Hierarchical cluster bootstrap) states that with $F = 4$ facilities the naive percentile cluster bootstrap is not, on its own, a frequentist-valid interval, and that the Ibragimov–Müller estimator is used as the primary inferential basis instead. The full justification: Cameron et al. [1] validate the wild cluster bootstrap from five clusters upward; MacKinnon and Webb [3] show it can fail with very few or badly unbalanced clusters; Canay et al. [2] prove a fixed-number-of-clusters over-rejection bound that, at $q = 4$ clusters, reaches $2^{1-q} = 0.125$ — up to 12.5 percentage points of over-rejection above the nominal level. This is why no conclusion in the main text rests on the cluster-bootstrap interval alone.

As a further check, we ran a Monte-Carlo calibration of the cluster-bootstrap’s own coverage under a known data-generating process matched to the real cluster sizes ($n_f = 45, 41, 20, 8$): empirical coverage was 0.994 against a 0.95 nominal target. The cluster bootstrap is therefore *over*-conservative here (valid but unnecessarily wide) rather than anti-conservative, so it is kept in the main text as an honest, if loose, descriptive comparison among resampling schemes, while the Ibragimov–Müller interval carries every claim about the exponent’s sign.

2 S2: Data provenance, full detail

Table 1 reproduces the main text’s Table “provenance” (Appendix A, Reproducibility) with the complete per-source extraction narrative; the main text keeps a condensed version.

3 S3: Adaptive-conformal ablation, extended

Section 6.1 of the main text (Discussion: ablations) summarizes the adaptive-conformal ablation in two sentences. Fig. 1 is the full figure: learned clusters (via unsupervised clustering on the covariates) simply rediscover facility identity, and the resulting prediction intervals are $\sim 25\%$ wider than normalized conformal’s with a LOGO-minimum of only 0.439 — *worse* than the facility-agnostic normalized-conformal baseline the main text recommends. This is reported as a negative

Table 1: Full data provenance by source (extended version of the main-text table).

Source	Reference (DOI)	n	Extraction detail	Redistribution
Vrancik1968	NASA TN D-4849 (1968)	41	All 41 points digitized from Fig. 13 (torque-coefficient vs. Reynolds-number curves for 4 rotor configurations); reading error estimated at 5–15% (mean 8.7%) from digitization-tool pixel resolution against the plotted axis range	Public domain (US Government report)
Xia2024	<i>Appl. Sci.</i> 14(17):7613; 10.3390/app14177613	45	37/45 points digitized from figures (windage-power-vs-speed curves across 5 rotor geometries); 8/45 taken from the paper’s reported CFD-validated closed-form fit rather than a plotted curve; err. 3–28.7% (mean 6.8%), highest for the lowest-speed points where absolute power is small and relative digitization error is amplified	CC-BY 4.0
Liu2024	Research Square preprint; 10.21203/rs.3.rs-4237491/v1	20	15/20 digitized from Fig. 7; 5/20 computed from the constant given in the paper’s Eq. 12; err. 5–10% (mean 6.3%)	CC-BY 4.0 (Research Square’s standard preprint license)
Zheng2024	<i>J. Zhejiang Univ.-SCI A</i> 25(11):938–952; 10.1631/jzus.A2300288	8	All 8 points are direct transcription of measured values from the source’s Table 1 (no digitization); the 1.4–83.1% range reported for this source is the <i>source authors’</i> own theoretical-model-vs-measurement discrepancy, not an uncertainty of our transcription, and is largest at the lowest rotational speed where windage power is smallest and relative discrepancy is correspondingly amplified	© publisher; derived numerical values only

result: letting the model additionally choose its own grouping does not help, and does not need to be re-derived to follow the main argument.

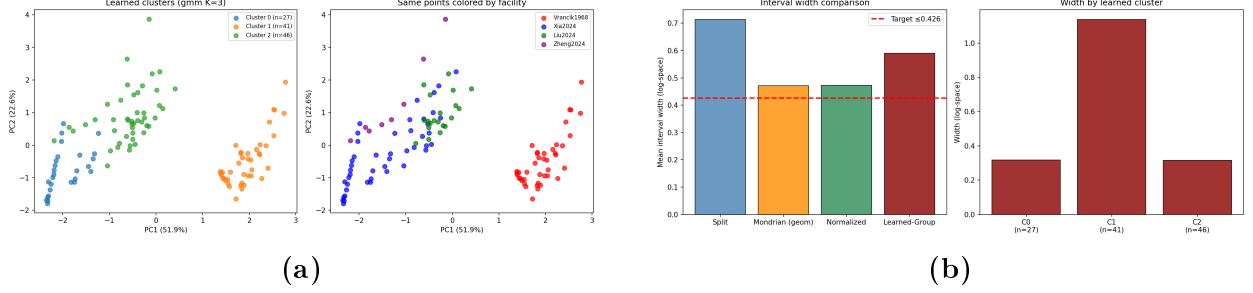


Figure 1: Adaptive-conformal ablation. **(a)** Clusters learned from the data (colour) overlaid on facility identity: the learned grouping rediscovers the facilities rather than a physical structure — the adaptation buys nothing. **(b)** Interval widths for the learned-group (adaptive) conformal variant versus the baselines: adaptation *widens* intervals ($\sim 25\%$) while failing the LOGO-minimum acceptance criterion (main-text Table “conformal”).

4 S4: Per-facility error in original units

The main text’s Table “baselines” reports $R_{\log}^2$ and R_{orig}^2 (aggregate, in-sample). Table 2 breaks the same in-sample fits down by facility, in original C_p units (MAPE and RMSE), computed from the identical fitted values underlying R_{orig}^2 — not a re-fit, not LOGO-CV (that is already a separate column in the main-text table).

Table 2: Per-facility MAPE (%) and RMSE (original C_p units) for each baseline in Table “baselines”, computed from the same in-sample fitted values as R_{orig}^2 . `scripts/facility_mape_rmse.py` in the released repository regenerates this table.

Model	Liu2024	Vrancik1968	Xia2024	Zheng2024	All
<i>MAPE (%)</i>					
Class-SR	6.9	22.5	5.9	4.8	11.9
C_p global power	44.2	33.7	82.1	33.2	54.6
C_p const. facility	7.6	51.0	61.9	6.7	44.6
C_p const. geometry	7.6	27.2	6.6	6.7	14.2
<i>RMSE (original C_p units)</i>					
Class-SR	0.0067	0.0426	0.0015	0.0050	0.0257
C_p global power	0.0406	0.0442	0.0086	0.0292	0.0329
C_p const. facility	0.0072	0.0703	0.0117	0.0070	0.0429
C_p const. geometry	0.0072	0.0474	0.0019	0.0070	0.0287

Two things stand out that the aggregate columns hide. First, Class-SR’s error is markedly uneven across facilities in original units (MAPE from 4.8% on Zheng2024 to 22.5% on Vrancik1968, a $4.7\times$ spread), consistent with the cross-facility heterogeneity the paper’s central claim describes rather than a uniformly good global fit. Second, C_p constant per facility (a model that literally memorizes each facility’s mean) is competitive with the global power law on Liu2024 and Zheng2024 but catastrophically worse on Xia2024 (61.9% MAPE) — a reminder that no single baseline is

uniformly informative across facilities, which is itself evidence against treating any one of them as “the” physical model.

5 S5: Naive/blind symbolic-regression replication of the S5 finding

A natural question about Section 5.5 of the main text (the S_5 case study) is whether an unbiased, off-the-shelf symbolic regression search independently rediscovers the same Mach-threshold structure, or whether finding it required deliberately including a regime-switching candidate in the hand-designed S_1 – S_6 family. We ran `gplearn`’s genetic-programming symbolic regressor (population 3,000, 40 generations, arithmetic primitive set only, no piecewise or threshold primitive available to it) on the same residual target the S_1 – S_6 family operates on ($\log C_p - \log C_i$, i.e. after removing the per-geometry prefactor already fit by class-SR), with $\log(\text{Re}_\Omega)$ and Mach as its only inputs and no access to facility or geometry labels.

Result: it does not. The fittest program after 40 generations is $-0.075 \times \log(\text{Re}_\Omega)$ (in-sample $R^2 = 0.470$) — a plain single-exponent power law that never incorporates Mach at all. An unconstrained genetic search, with no regime-switching primitive handed to it and no hint that a threshold structure might be worth trying, converges to the simplest fit and does not stumble onto the two-regime artifact.

We read this as a genuine scope clarification, not a weakness the main text should downplay: the S_5 vulnerability specifically requires a structure-selection search whose *candidate library* already includes regime-switching forms (as ours does, and as increasingly capable automated hypothesis-generation pipelines – the kind this protocol is meant to audit – routinely do), not every symbolic-regression pipeline regardless of search-space design. A minimal, generic evolutionary search with a narrow primitive set will not independently reproduce this failure mode; a Bayesian search over an expanded candidate family that includes threshold/regime structures can. `scripts/naive_sr_replication.py` in the released repository reproduces this result exactly.

6 S6: Synthetic power curve, structural (not prefactor) confounding

The main text’s specificity/power analysis (Section 5.6, Fig. “specificity_power_curve”) injects a *prefactor*-level confound (a constant additive offset on one facility’s $\log C_p$) and reports low detection power by design, because class-SR’s per-geometry prefactor absorbs it. Here we inject a *structural* confound instead: one facility’s Reynolds exponent itself is shifted by Δq relative to the other three, and — matching the actual mechanism behind S_5 , where Liu2024’s distinct behaviour coincides with it occupying its own Mach range — that facility’s Mach values are also drawn from a disjoint range, so the exponent shift is correlated with Mach exactly as the S_1 – S_6 Bayesian search could pick up on. (An earlier version of this experiment gave the confounded facility the same random Mach range as everyone else; since the S_1 – S_6 family only searches for regimes defined *on Mach*, an exponent shift uncorrelated with Mach gave it nothing to find, and measured power sat at the false-positive floor for every Δq – a bug in the experiment, not a finding about the protocol, corrected before the results below.)

Two things are worth noting honestly. First, power at $\Delta q = 0$ (no real exponent difference at all) is already 0.13, not 0: the confounded facility’s disjoint Mach range alone – with no genuine structural difference – is enough to occasionally trigger an “ambiguous” or “facility-artifact” verdict, since Test 1 (threshold-vs-facility contingency) reacts to the covariate separation itself. This is consistent with Test 1 being, on its own, a qualitative flag rather than a stand-alone significance gate

Table 3: Detection power (fraction of $K = 100$ synthetic datasets flagged facility-artifact / statistically-real-spurious / report- with-disclaimer) as the injected exponent gap Δq grows, with the confounded facility’s Mach range disjoint from the other three. `scripts/structural_confound_power.py` in the released repository regenerates this table.

Δq	Power	95% CI (Clopper–Pearson)	Dominant verdict
0.00	0.13	[0.07, 0.21]	no-regime-detected (87/100)
0.02	0.17	[0.10, 0.26]	no-regime-detected (83/100)
0.05	0.16	[0.09, 0.25]	no-regime-detected (84/100)
0.10	0.30	[0.21, 0.40]	no-regime-detected (70/100)
0.20	0.70	[0.60, 0.79]	ambiguous (70/100)

(Section 5.5 of the main text). Second, the dominant verdict at the largest gap tested ($\Delta q = 0.2$) is *ambiguous* rather than *facility-artifact* outright: the protocol increasingly refuses to call a clean verdict as the confound grows, rather than confidently misclassifying it as physics, which is the behaviour the protocol is meant to have in a genuinely uncertain regime.

7 S7: Pairwise covariate-overlap (MMD)

Section 3.1 of the main text (Problem formalization) quantifies the extreme covariate shift across facilities with the scalar characteristic-scale ratio $\Lambda = 70.4$. Table 4 complements this with a direct, multivariate quantification: the unbiased, RBF-kernel maximum mean discrepancy (median-heuristic bandwidth) between every pair of facilities on the joint $(\log \text{Re}_\Omega, \text{Ma})$ covariate space, each standardized to zero mean/unit variance before computing the kernel.

Table 4: Pairwise squared MMD on standardized $(\log \text{Re}_\Omega, \text{Ma})$. Larger values indicate less covariate overlap. Facility sample sizes in parentheses.

Facility pair	n_1, n_2	MMD ²
Liu2024 vs. Vrancik1968	20, 41	0.101
Liu2024 vs. Zheng2024	20, 8	0.125
Vrancik1968 vs. Zheng2024	41, 8	0.158
Xia2024 vs. Zheng2024	45, 8	0.634
Liu2024 vs. Xia2024	20, 45	0.757
Vrancik1968 vs. Xia2024	41, 45	0.913

All six pairs show non-trivial separation; none is close to 0 (complete overlap). The ordering does not simply track the univariate Mach-threshold story of Section 5.5 (Liu2024 is not the most separated facility in this joint space; Vrancik1968 and Xia2024 are) — a reminder that the S_5 discriminant battery targets one specific, Mach-conditioned mechanism, not overall covariate distance, and that a facility pair can be jointly far apart in (Re, Ma) space without that separation aligning with any structure a candidate family searches for.

References

- [1] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008.

- [2] Ivan A. Canay, Andres Santos, and Azeem M. Shaikh. The wild bootstrap with a small number of large clusters. *The Review of Economics and Statistics*, 103(2):346–363, 2021.
- [3] James G. MacKinnon and Matthew D. Webb. Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2):233–254, 2017.